AI-Assisted Analysis of Secondary Battery Technology

1. Research Topic

AI-assisted analysis and management of secondary batteries (lithium-ion batteries) focusing on performance, safety, and lifetime prediction

2. Motivation and Purpose of Research

With the rapid expansion of electric vehicles (EVs), energy storage systems (ESS), and renewable energy infrastructure, secondary batteries have become a critical component of modern society. However, practical problems such as battery fires, inaccurate lifetime prediction, and inefficient energy management continue to limit their reliability and safety.

The purpose of this research is to address these real-world issues by analyzing how Artificial Intelligence (AI) can be applied to secondary battery technology. This study aims to demonstrate that AI-assisted analysis can improve battery safety, performance prediction accuracy, and operational efficiency. As a student majoring in electrical engineering, this research also seeks to explore the practical integration of AI into future power and energy systems.

3. Research Methodology Using AI

This study was conducted using an AI-assisted research framework, in which AI functioned as an active research tool rather than a passive reference source.

The research process consisted of the following steps: - AI-based data collection from technical papers, industry reports, and battery research articles - AI-assisted summarization and classification of battery-related concepts - Comparative analysis of traditional battery management methods and AI-based approaches - AI-supported interpretation of battery performance, degradation behavior, and safety mechanisms - Refinement of technical explanations and English expressions using generative AI

Through this structured methodology, complex battery technologies were analyzed efficiently and systematically.

4. Overview of Secondary Batteries

Secondary batteries are rechargeable energy storage devices that convert electrical energy into chemical energy during charging and reverse the process during discharging. Among them, lithium-ion batteries are the most widely used due to their high energy density and long cycle life.

4.1 Basic Operating Principle

- Charging: Electrical energy is stored as chemical energy through electrochemical reactions.
- Discharging: Chemical energy is converted back into electrical energy to power electrical devices.

4.2 Components of Lithium-Ion Batteries

- Cathode: Lithium metal oxide
- Anode: Graphite
- Electrolyte: Medium that enables lithium-ion transport
- Separator: Insulating membrane preventing short circuits

5. Overview of AI Technology

Artificial Intelligence (AI) refers to computational techniques that learn patterns from data and generate predictions or decisions. In battery research, machine learning and deep learning are widely used to analyze large-scale operational data.

5.1 Machine Learning

Machine learning models analyze historical battery data to predict future behavior such as capacity degradation and voltage variation.

5.2 Deep Learning

Deep learning models utilize neural networks to process complex, non-linear battery data, enabling high-accuracy prediction and anomaly detection.

6. Applications of AI in Secondary Batteries

6.1 Battery State Estimation (SoC and SoH)

AI-based models analyze voltage, current, and temperature data to estimate the State of Charge (SoC) and State of Health (SoH). Previous studies report that AI-based estimation methods reduce prediction error compared to traditional model-based approaches, enabling more accurate real-time monitoring.

6.2 Battery Lifetime Prediction

By learning long-term charging and discharging patterns, AI can predict battery degradation trends and remaining useful life. This contributes to optimized maintenance scheduling and cost reduction.

6.3 Safety Enhancement

AI systems can detect early warning signs of abnormal behavior such as thermal runaway, overcharging, or internal short circuits. Early detection significantly reduces the risk of battery fires and explosions.

6.4 Manufacturing and Quality Control Optimization

During battery production, AI analyzes defect patterns and process data to improve product consistency and manufacturing efficiency.

7. Relevance to Electrical Engineering

AI-assisted secondary battery technology is directly connected to electrical engineering fields such as power electronics, control systems, and automation. In particular, Battery Management Systems (BMS) integrate electrical measurements with AI-based algorithms to ensure safe and efficient battery operation.

This research demonstrates how electrical engineering knowledge can be expanded through AI-based analysis, highlighting the importance of interdisciplinary competence.

8. Expected Effects and Future Outlook

- Improved battery safety and reliability
- Enhanced prediction accuracy for battery lifespan
- Reduced operational and maintenance costs
- Increased demand for engineers with combined expertise in electrical engineering and AI

AI-assisted research is expected to become a standard approach in future battery development and energy system optimization.

9. Conclusion

This study was conducted as part of an AI utilization contest and presents an AI-assisted research model for analyzing secondary battery technology. By integrating AI into the entire research process, this study demonstrates that AI can significantly enhance the efficiency, accuracy, and clarity of technical analysis.

The findings confirm that AI-assisted research is a reliable and practical approach for addressing real-world battery challenges. This research highlights the potential of human–AI collaborative research as an essential methodology for future engineers in the energy and electrical engineering fields.